

Nike – Enterprise Cloud Modernization & Azure to AWS Migration

Executive Summary

Nike initiated an enterprise-scale cloud modernization program to migrate analytics platforms, operational databases, and containerized applications from Microsoft Azure to Amazon Web Services (AWS). The objective was to improve scalability, governance, operational efficiency, performance, and cost optimization while supporting rapidly growing retail, e-commerce, and supply chain analytics workloads.

Business Challenges

- Rising infrastructure costs across Azure Databricks, ADLS Gen2, PostgreSQL, and AKS workloads.
- Increasing data volumes generated from stores, digital commerce, inventory, and supply chain operations.
- Fragmented governance and limited visibility into data lineage and access management.
- Complex dependencies across ETL pipelines, databases, and microservices.
- Need for a cloud-native architecture capable of supporting future AI and advanced analytics initiatives.

Source Environment

Azure Databricks for analytics workloads, ADLS Gen2 as the enterprise data lake, Azure PostgreSQL databases for operational applications, Azure Kubernetes Service (AKS) for microservices, Azure Data Factory for orchestration, Synapse Analytics for reporting, and Azure Key Vault for secrets management.

Migration Flow

1. Assess existing Azure Databricks workloads, PostgreSQL databases, AKS applications, and associated dependencies.
2. Migrate historical data from ADLS Gen2 to Amazon S3 using AWS DataSync.
3. Replicate and migrate Azure PostgreSQL databases to Amazon Aurora PostgreSQL using AWS DMS.
4. Replatform AKS-based microservices and APIs onto Amazon EKS clusters.
5. Migrate Azure Databricks notebooks, jobs, clusters, and Delta workloads to AWS Databricks.
6. Rebuild orchestration workflows using AWS Glue and Amazon MWAA (Apache Airflow).
7. Implement Unity Catalog and AWS Lake Formation for centralized governance and access control.
8. Execute automated row-level validation, reconciliation, and performance testing.
9. Cut over reporting, dashboards, and downstream applications to the AWS platform.
10. Optimize compute, storage, monitoring, and security configurations for long-term operations.

Target AWS Architecture

Data Platform: AWS Databricks, Delta Lake, Amazon S3, Unity Catalog.

Database Layer: Amazon Aurora PostgreSQL.

Container Platform: Amazon EKS.

Data Integration & Orchestration: AWS Glue, AWS DMS, Amazon AppFlow, Amazon MWAA.

Governance & Security: AWS Lake Formation, AWS IAM, AWS KMS.

Monitoring: Amazon CloudWatch.

Analytics & Consumption: Tableau, Power BI, Databricks AI/BI Dashboards.
Advanced Analytics & ML: Amazon SageMaker.

Migration Validation Framework

A comprehensive validation framework was implemented to ensure migration accuracy. The framework performed row-count reconciliation, schema validation, business-rule checks, data quality monitoring, and performance benchmarking across both Azure and AWS environments. This approach ensured minimal business disruption and maintained data consistency throughout the migration.

Business Value Achieved

- 20–25% reduction in cloud infrastructure costs through storage and compute optimization.
- 35% improvement in pipeline execution performance.
- Centralized governance and metadata management using Unity Catalog.
- Improved scalability for global retail, e-commerce, and supply chain workloads.
- Reduced operational overhead through automation and standardized processes.

Impact Generated

Operational Efficiency: Reduced manual intervention through automated migration, validation, and orchestration processes.

Performance & Scalability: Enhanced workload performance and scalability using AWS Databricks, Aurora PostgreSQL, and Amazon EKS.

Data Reliability: Achieved greater than 99.9% validation accuracy through automated reconciliation frameworks.

Business Agility: Accelerated onboarding of new datasets, applications, and analytics use cases across business functions.

Technology Stack

AWS Databricks, Amazon S3, Delta Lake, Unity Catalog, Amazon Aurora PostgreSQL, Amazon EKS, AWS Glue, AWS DMS, Amazon AppFlow, Amazon MWAA, AWS Lake Formation, AWS IAM, AWS KMS, Amazon CloudWatch, Tableau, Power BI, Amazon SageMaker.

Conclusion

The migration established a secure, scalable, and cloud-native AWS data platform that supports Nike's analytics, operational applications, and future AI initiatives. By modernizing data, database, and container workloads, Nike achieved lower costs, improved performance, stronger governance, and a future-ready foundation for innovation and growth.